CHAPTER 4

FROM SUPERSTITION TO SCIENCE

THE CHALLENGES WE FACE TODAY CANNOT BE RESOLVED by antiquated notions and values that are no longer relevant. Unfortunately we tend to support the basic values and traditions that reflect the past, without questioning their appropriateness to the present or the future. The more superficial the changes, the more things remain the same. For us to think creatively about our future and examine our traditional habits of thought, we must become better informed. We must look at other alternatives objectively rather than trying to fit the future into our present social mold.

Even today, millions of people throughout the civilized world still worship many different gods and fear demons, while others tend to placate their gods with incantations, sacrifices, adulation and flattery. Still others use astrological charts and pendulums for arriving at decisions. Even many of the most popular newspapers feature a column on astrology, while television and radio airwaves are filled with psychic problem solvers. A noted psychic recently proclaimed we would be surprised at just how many of the important decisions of running our country were in the hands of soothsayers and charlatans.

Until the discipline of scientific inquiry came of age, human beings could not comprehend their relationship to the physical world, so they invented their own explanations. These explanations tended to be simple and some harmful resulting in religious rituals superstitions, astrology, numerology, fortune telling, etc. Millions of people still accept and follow these ancient beliefs.

It is not that scientists are closed-minded regarding these issues, their standards for accepting such ideas simply require a much more rigorous and sophisticated method of inquiry. The difference between the scientist and the metaphysician is that the scientist asks a question and engages in experiments to determine the nature of the physical world; part of this process requires that the experiments be verified by others who must get the same result. In contrast, the metaphysicians fabricate answers that are pleasing to them and which require no verification, a process involving little or no effort.

When we consider the extent to which metaphysicians rely on this unverifiable information for direction, it is ironic to see them immediately surrender their lofty intuitive and spiritual interpretations of worldly things when it comes to their daily lives. When

purchasing property, for example, they measure exactly how many square feet are being exchanged for a given sum of money. When purchasing a new automobile, they ask how many miles per gallon the car will deliver, or the precise amount of fuel needed to travel a given distance.

In fact, most of our daily living is based upon the application of scientific principles. As B.F. Skinner would say, "intuitive feelings may tickle the cockles of a poet's heart", but this does nothing to enhance our knowledge of the physical world. What makes a person feel good or appeals to one's emotions does not necessarily add to one's understanding of our world.

Throughout history life for most people has been a perpetual struggle against a great many problems: finances, health, personal safety, security, starvation, and a myriad of others. Failing to find a safe haven in a world in which many are resigned to the consequences of our original sin, theologians created the concept of a distant Heaven, a place of eternal bliss and a limitless abundance of warmth and love, where all of the people are free of destitution, greed, lust, the need for money and all the other afflictions that plagued humankind for centuries.

Of course, to qualify for entrance into this world of eternal bliss, one must die and, prior to that, demonstrate impeccable earthly behavior, engage in constant prayer to an intermediary for forgiveness for one's transgressions.

Others seek to attain this end while still on an earthly plane through meditation or renunciation of the material world. By this means they claim they will be able to experience Nirvana. Although it is true this form of meditation will alter their associative memory and develop the necessary procedure for fulfilling their hopes, dreams and wishes, the attainment of this state takes place only in their own mind. This procedure for inducing wish fulfillment and dreams of unique individual fantasies often makes it extremely difficult for people to know the difference between the physical world and their created fantasies.

People will continue to search for answers to these universal and perplexing problems. But to find meaningful answers, one must first know the appropriate questions to ask. People pose questions far too complex for them to unravel, without their first processing the fundamental knowledge of what it is they are seeking.

Even in the world of science, which is a closer approximation of the physical world, it is acknowledged that there are no absolutes. If science were to accept absolutes, the field of scientific inquiry would come to an end.

There are those who are in search of the truth. This is an endless search that will take one nowhere. If we ever find out exactly who we are, it will be the end of the human intellect. Whether they realize it or not, most people continue to undergo change in their values, outlook and understanding. This process has no finality. Human beings are constantly evolving organisms. The question that remains is: how does one select, of the very many alternatives, that which is relevant?

A brief course in preliminary scientific principles enables a person to better understand the world they live in and their relationship to it. We can only view the world with our receptors and the degree of linguistic precision that we acquire in our culture. No human being can view anything with the certain knowledge that they are perceiving it as it really is. For instance, if a mouse could write it would describe the dog as an enormous creature; whereas the giraffe would take issue with that and say it is a tiny creature from his elevated point of view. They are both telling the "truth" as they see it but only from their limited point of view.

Other questions are asked such as "What is the meaning of life?" "What is consciousness and the mysteries of the mind?" "Why am I here?" "What is my relationship to God and the universe?" These same questions have been asked over the centuries, and most are totally irrelevant to achieving social progress and, eventually parity. These are examples of unanswerable questions because none have a valid point of reference. Does the posing of such ambiguous questions express concern over the condition of one's fellow human beings, or a desire to elevate their condition? Of course not. Such musings are pure gibberish, as impotent as wailing over an injured person rather than seeking medical attention.

Take, for example, the question "what is life, its meaning and our relationship to the universe?" an ultimately hollow and meaningless question. There is no effort put out by these philosophers, poets and metaphysicians for the genuine pursuit of the processes in question. They understand so little about the mechanisms of the physical processes of nature, yet those asking such questions don't go into the lab in pursuit of this process or try to understand even the structure of a single cell, let alone the universe. They are merely repeating the quotations other verbalists have made for centuries, while making no effort to verify the validity of their assumptions. Although they feel these questions are profound, in this context, they are actually naive.

Besides, with such questions about the process we call life we make an assumption that life has "meaning." Actually, as difficult as it may for some to accept, the only meaning it has is that which we give it. Genuine concern for understanding such profound questions is manifested by engaging in a research program where examine the characteristics and mechanics of living systems. This is precisely the same analytic principle by which an increase in criminal behavior requires an investigation into the nature and factors that shape human behavior.

Merely talking about the things that we do not understand does not add to our knowledge. For example the word "instinct" does not tell us anything more about the behavior of the organism involved. It is a word assigned to patterns of behavior many do not understand. Instead of the word instinct, we need precise information on the actual processes behind such things as how fish migrate, how birds build nests, and how organisms undergo adaptation.

One may reasonably ask: Why do people cling to the values of the past, when they seem so obviously irrelevant to the present? Part of this, of course, is that long-standing

thought patterns are often hard to overcome — particularly when they appear to serve the interests of the individual. The old way of thinking is simpler and easier to handle. In a two-valued way of thinking, such as good and bad, right and wrong, love and hate, cause and effect, very little analytic process is involved.

Also, many of us are ill equipped in analytical thinking. Analytical thinking requires an understanding of process relationships and a broad range of information. We are insufficiently equipped and trained to objectively evaluate alternative proposals. Science today is taught as if it were a series of discrete specialties, as if biology and chemistry and physics ere not really a single science. No school we know of presents science in a holistic and significant way. Students learn narrow principles and laws and processes, rather than the scientific way of thinking. This makes it very difficult for the average person to apply scientific and analytical thinking in their everyday lives.

That is the main reason ignorance prevails. People want answers instantly, ones they can easily grasp and use immediately — even ones with no basis in fact. Science does not provide quick ultimate answers. It merely presents information about the physical world we live in. The scientific community recommends a system to best explain how it is that nature functions the way it does.

The challenge scientists must confront in the near future is to develop methods of presenting science and technology in a language easily assimilated by those less familiar with the scientific method. This may be accomplished through films, books, videos and CD's, to help bridge the difference between science and irrelevance. At present most of these difficulties lie in the field of communication and education. We recommend *The Demon-Haunted world: Science as a Candle in The Dark*, by Carl Sagan to anyone interested in exploring this area.

The reason we emphasize science is that science-oriented individuals present their findings without any regard as to whether people like it. Often at the risk of their social standing, careers, or even their lives, they hold fast to such concepts as the earth being neither flat nor the center of the universe, the theory of evolution, and the fact that illness is not due to punishment by gods or demons. This differs vastly from politicians who seek public approbation by catering to the dominant values of the times. We see examples of this in such emotional populist issues as family values, nationalism, and religion.

Most churches make people feel guilty about natural human inclinations, making them dependent on the church for forgiveness. While religion caters to unresolved human problems such as insecurity, shame, fear, wish fulfillment, and offers hope for a better life if not in this world then the next, science offers people the tools of reason and knowledge to help build self-reliance and free people from the mythology of wish fulfillment.

All human beings have the potential to develop their own concepts, and to make their own — heaven or hell here on earth. But there is no way for these refugees from reality to perceive the actual state of affairs without putting out tremendous effort and inquiry to translate their wishes and dreams into a working reality. It takes this honest effort to understand the nature of the world we live in.

Should people turn to science for answers, most are not sophisticated enough to state the problems correctly, let alone grasp the response to their inquiries. Following the path of least resistance in our thinking, however, only holds us back from making a more appropriate evaluation of any area of our investigation. This makes it easier to understand how dictators such as Hitler succeed in building a large following, particularly during hard times.

In seeking simplified answers they blame many social problems on minorities, foreigners, karma, auras that are believed to surround each individual, acts of demons or gods, or the position of the planets at the time of one's birth, just to name a few. Others seek to bring out higher levels of human consciousness and self-realization through meditation. To the uninformed these are easy to understand, because they do not demand concrete proof or verifiable evidence. Hence the popularity of metaphysics. Some insist that insist we return to the simpler life of the past, to the "good old days." This is another myth that some people cling to, that things were somehow "better" in times of less technological development.

This is unfortunately a growing phenomenon in the scientifically illiterate world. Even some scientists today are persuaded by pseudo-science. In a sense, even scientists are victims of culture. The proof of this lies in the fact that some have been persuaded to use their abilities to make weapons of destruction with little thought to the consequences of their actions.

The belief, also, that science or religion would not conform to the indoctrination of totalitarianism is a continuing myth. At various times in recent history, in countries such as Spain, Italy, Russia, Japan and Germany, areas of science gave way to sadism and even the practitioners of one of the most ethical professions — medicine — performed gruesome experiments on living people. Many churches of nations at war could be found blessing tanks, soldiers and battleships ships, even when the combatants on both sides were members of the same denomination.

There is really no such thing as a pure scientist, as all data is interpreted on the basis of personal background and experience. Some people may be said to be scientific in their specialized discipline while in other areas of physical science they may be illiterate. Actually formulating conclusions outside of one's particular discipline would be a violation of the scientific method.

Science should not be used for the conquest of nature, but rather to point out our interdependence and connectivity to nature, and to explore how to utilize our knowledge to live in accordance with the natural order of things. When we as a nation spend nearly three hundred billion dollars annually on defense and only two billion towards understanding our environment, one must question whether there is actually intelligent life on Earth.

The only hope for the development of a new civilization for humankind is to accept the responsibility for improving our lives through knowledge, understanding and a deeper comprehension of humanity's relationship to natural processes of evolution. Our future is determined by the effort that we put forth to achieve this transition.

Furthermore, when we finally outgrow assumptions of superior and inferior races and realize the unity of humankind and its true relationship with the planet, we will achieve the full potentials of science for humane development. This could serve as a unifying, global force for the achievement of a sustainable world.

But not knowing where we are, how can we possibly know where we are headed?

CHAPTER 5

NEW FRONTIERS OF SOCIAL CHANGE

IN OUR DYNAMIC UNIVERSE ALL THINGS CHANGE, from the state of the farthest reaches of outer space to the movement of the continents. The same process of change occurs in all living and nonliving systems. The history of civilization is the story of change from the simple to the more complex. Human ingenuity and invention bear witness to this fact. No system can remain static for long. Unfortunately, the changes are not always for the best.

Although we accept the inevitability of change, humans meet change with considerable resistance. Those in charge, whether of a religious, military, socialist, capitalist, communist, or tribal institution or society, will attempt to hold back change, because change threatens those in control. Even those oppressed by a social system may support it and the status quo because it is familiar and known. No matter how oppressive one's surroundings, there is comfort in the familiar.

Human civilization is no exception to this process of change. Change occurs to all social systems, and is the only constant. The history of humankind is one of change, which is either brought about by natural circumstance or by the direct intervention of human effort.

Technology influences the most remote regions of the world almost as fast as the technology develops. In 1993 Malaysia had interests in banking, construction, issuing of credit cards, fast food outlets, manufacturing of medical supplies, and information technologies.

Untouched, isolated cultures are fast becoming the stuff of history. Although many native peoples still wear ancestral dress, they also carry video cameras and other state-of-the-art electronic devices. These invasions of newer technologies are present from Papua New Guinea to Vietnam to China. In Thailand, we find Siam Cement, one of the largest cement companies in the world. Some of the most successful cement companies in the world can also be found in Colombia and Peru. In our own country, Disney Information has replaced US Steel in digitally transmitted information.

Yet at every turn, vested interests oppose technological change. Earlier this century defenders of the horse-mounted cavalry delayed development of the tank; so entrenched

was this tradition that when Germany invaded Poland in 1939, their Panzers faced Polish soldiers still mounted on horseback.

It was immediately and fatally obvious that the tank had rendered the horse soldiers obsolete. Later the development of aircraft threatened the tank divisions. Pilots and aircraft designers fought to hold back the development of guided missiles. The missile men fought to hold back the development of laser weapons.

In a similar manner the established social order tends to perpetuate itself. Those in positions of power are able and highly motivated to delay conditions that would advance society as a whole.

From the introduction of agriculture some 10,000 years ago until recent history — specifically, the advent of the Machine Age in the late seventeenth and early eighteenth centuries — the rate of this change was slow. Social change in times past crept along, accompanied by a great deal of human suffering during the transition from one phase of civilization to another. Since the Industrial Revolution, change continues to accelerate at a fantastic rate. In technologically advanced cultures, change occurs rapidly — often too rapidly for the average person to comprehend or adjust to. And even when individual adjust, our institutions of — government, education, medicine and industrial cannot. Their size, their infrastructure, their very processes and missions resist and oppose rapid change.

In just a few decades the transfer of information moved from the telegraph to radio to television to wireless transmission computers to satellites that can store trillions of bits of data and transmit this to any part of the globe instantaneously.

Many of us forget or are not aware of the fact that less than forty years ago it took a pair of wires to carry a dozen conversations. Twenty years later, one cable conveyed thirty thousand conversations simultaneously. Today a single laser beam carries more than a million. This explosion of technology can no longer be stopped.

Whether the citizenry of the world is capable of grasping the significance of such prodigious changes is actually irrelevant. What is required and really significant is that a sufficient number of the world's leaders be able to comprehend developments of this magnitude. The degree to which we are able to comprehend such developments will determine our chances of survival.

Technological changes occur less rapidly in lesser-developed countries. Many of the systems and methodologies of some nations have been remained for hundreds, even thousands, of years. Small groups of people, such as the headhunters of the Amazon, still live in places where their social and physical environment remains relatively static. They can still be found making the same kinds of rafts and other tools, and utilizing the same techniques as their ancestors did a thousand years ago.

Stagnation is not confined to the underdeveloped countries; even today among developed countries there are large national groups of people who cling tenaciously to the past as the benefits of civilization pass them by. But the future is no respecter of the dominant

values of today. And the generations to come will evolve a set of values unique to that phase of civilization.

While behavior patterns may have remained unchanged for thousands of years in these low-technology societies, the notion that people may be carbon copies of their predecessors is untrue in today's technological world. New generations immersed in different environments require different solutions.

With the advent of the World Wide Web, cybernation and artificial intelligence, the rate of change is being greatly accelerated. Possibly in the next ten years we will see more changes than those wrought by the past events of recorded history. If we as a nation fail to adapt to these changes others will pass us by. The future belongs to those capable of meeting these challenges.

To be sure, technology evolves at its own rate; one thing triggers another, leading to wider applications. Future technologies evolve at their own pace, determined by many interrelated factors. If we attempt to alter our social evolution faster than the ability of society to adapt to change, there will be severe consequences. Rapid changes without the proper preparations generate severe problems. The development of social systems unresponsive to people and their environment only increases our internal strife.

Common crises create common bonds. While people seek advantage during the good times, shared suffering draws people together. We have seen this behavior repeated time and time again throughout the centuries, during times of flood, famine, fire, or natural disasters. When the threat passes, people return to seeking advantage over others.

Motion pictures like *Independence Day* depict a world united to repel an invasion by a superior, hostile alien culture. Indeed, it seems that the only force that would mobilize the world in a unified direction would be one that poses a common threat, such as a huge meteor heading towards the earth or some other major catastrophe. In such an event, border disputes would cease and become irrelevant to the impending disaster. While many would call upon their deities to intervene, nations would unite and call upon science and technology to help solve a common threat. Bankers and lawyers, businessmen and politicians would be bypassed. *All of the resources for total mobilization would be harnessed without any concern for monetary cost or profit.* Under this kind of threatening condition, most people realize where their survival lies.

As global challenges and scientific information proliferate, we face common threats that transcend national boundaries: overpopulation, energy shortages, environmental pollution, water scarcity, economic catastrophe, the spread of uncontrollable diseases and the technological displacement of people by machines threaten us all. Although many people are dedicated to alleviating these problems, our social and environmental problems will remain insurmountable as long as a few powerful nations and financial interests maintain control of and consume most of the world's resources.

Although people, publications, and multi-media presentations paint spectacular pictures of the developments to come in such areas as transportation, housing, and medicine,

they ignore the fact that, in a monetary-based economy, the full benefits of these developments continue to be distributed to a relative few. What is not touched upon is how the new technologies of the future can be used to organize societies and economies efficiently and equitably so that everyone may benefit from them.

Currently, no think tanks conduct brainstorming sessions how to align social organization with the advancements of technology. No government or industrial group plans for the displacement of people by machines.

Many people believe that in the event of social breakdown the government will ensure their survival. This is highly improbable. In the event of such a breakdown, the existing government would most likely declare a state of emergency in an attempt to prevent total chaos. A review of actions taken by governments facing social collapse over the last few decades shows that their primary concern is preserving existing institutions and power structures — even though these may be a chief contributing factor to the problem.

Many people throughout history have taken politicians to task for actions that have not been on the people's behalf. The reasons for this may become clearer when one accepts the fact that even in modern democracies, these leaders are not elected to improve the life of the average person: rather, they are appointed to maintain the preferential positions of many of the established order.

There are growing indications of awareness on the part of people in various areas of the world that events have gone beyond the control of political leaders. Everywhere we can see political figures and parties come and go, political strategies adopted and discarded for their inability to satisfy the demands of one faction or another.

The reason we do not stress writing your congressman, or any number of governmental agencies, is that they lack the necessary knowledge to deal with our problems. Their focus is on preserving existing systems, not on changing them. It appears there are few within the present day societies who want to phase themselves out. In modern industrial societies the cause of inaction lies within the cumbersome political process itself, an anachronism in an era when most decisions can be made on any important issue in a split second by the objective entry of data into computers.

Real social change occurs when conditions deteriorate to such an extent that governments, politicians, and social institutions no longer have the support and confidence of the people. What once worked is acknowledged as no longer relevant or acceptable. When the public is better informed or in sufficient pain, only then is it possible to introduce a new social arrangement.

Unfortunately, the majority of people today respond to simplistic answers, which tends to repeat the cycle of events. When faced with intolerable social conditions, many of the older patterns will emerge again as people attempt to find someone or something to blame for the conditions (blacks, Jews, or homosexuals, for instance) or as they seek refuge in religion or a belief in supernatural forces.

Significant social changes are not brought about by men and women of reason and good will on a personal level. The notion that one can sit and talk to individuals and alter their values is highly improbable. If the person one is talking to does not have the fundamental knowledge of the operation of scientific principles and the processes of natural laws, it is difficult for them to understand how things fit together on a holistic level.

The solutions to our problems will not come about through the application of reason or logic. We do not live in a reasonable or logical world. There is no historical record of any society that deliberately and consciously modified their culture to fit changing times. The real factors responsible for social change result from biosocial pressures inherent in all social systems. It is brought about by natural or economic occurrences which immediately threaten large numbers of people.

Some of these conditions responsible for social change include limited resources, war, overpopulation, epidemics, natural disasters, economic recession, downsizing on a mass scale, technological displacement of people by machines and the failure of their appointed leaders to overcome these problems.

Change can come from disasters or from major technological advances. The introduction of agriculture brought about a significant change in society, as did the Industrial Revolution and the introduction of the medium of money to the exchange process. From an historical perspective all of these appear positive. At the time of their inception, however, people lost their jobs, new skills were required, whole ways of life disappeared.

The direction change takes is not always for the better or for the improvement of the human condition. Change is risky when scarcity, artificial or real, drives the economy and power-seeking leaders command weapons powerful enough to annihilate entire populations and render our planet uninhabitable. Humankind's potential for creativity and innovation far exceeds its inclination to destroy — yet every time that we exercise destructive power, we take a thousands steps backward for every few forward.

Certainly, not all change has been beneficial to humanity or to the integrity of the planet's life support systems. For this reason, many people desire a return to earlier and simpler times. But it has been demonstrated that any effective, large-scale and permanent social transformation cannot be achieved through the development of small, cooperative and hand-tooled economies. These cooperative ventures have been tried throughout history by both religious and secular interests. Most failed to achieve or sustain their desired goals. The reason for the failure was not due to human nature or greed. The primary cause of their demise was the fact that most participants, although sincere, had little information on the primary factors responsible for human behavior.

Although individuals throughout history have proposed many ideal social arrangements — from Plato's Republic through the modern Utopians — no industrial nation have ever proposed an arrangement to improve the lives of all people and build a truly civilized nation. This is not difficult to understand when one considers the overall principles that

govern the social system. The government's principles are based on ownership and the accumulation of wealth, power and property.

Visionaries of sincere intent write and speak eloquently of a world moving forward in unity and brotherhood. Many expect a worldwide epiphany or transformational event. Others expect reason to prevail. Only a very few propose plans to achieve unity, some of which appear incomplete and pose a threat to existing institutions and to national and self-interest: the architects of such plans are often classified as agitators, impractical utopians, and disrupters.

A few bold attempts to achieve world unification failed because movement leaders had no real understanding of the primary forces that shape social evolution. Most significantly, they sought their solutions within the framework of the monetary system, never realizing that physical resources — not money — are the real necessity where the ability of any social system to sustain its people is concerned.

Although the monetary system served to help to eliminate the old, cumbersome rates of exchange in a barter system, it is by no means the final answer. The history of civilization indicates continuing evolution and adaptation. No single answer is able to serve all time, all peoples, and all problems.

Our outmoded social, political, and international order is no longer appropriate to these times. Our outworn social institutions cannot grasp or adapt to the ability of innovative technology to achieve good, nor can they overcome the inequities forced upon so many.

Competition and scarcity have instilled an atmosphere of jealousy and mistrust within people and nations. The concepts of proprietary rights, manifested in the corporate entity and in the sovereignty of nations, inhibit the free exchange of information that is necessary to meet global challenges.

Many people fee the pain of change and yearn to return to a simpler time of "traditional" values. Their vision is faulty. Those times were not, in fact, so good. In the first fifty years of that "simpler" time, we waged two world wars. In the intervening years a major agricultural and economic disaster sent millions to soup kitchens and breadlines. If they are honest, it is not the fantasy of the "good old days" that they wish to see realized, but more simplicity that they want.

Our problems today are enormous and global in their scope and impact. They cannot be solved by any one nation. The concept of common good is global in nature, local in implementation. We cannot hope to backtrack to traditional values, which no longer apply. Any attempt to retreat into the methods of the past would condemn untold millions to a life of needless misery, toil, and suffering.

I am not advocating that these older institutions be overthrown: it is just that they are becoming unworkable. Unfortunately, it will most likely take a social and economic breakdown to bring about the demise of the old system and its institutions. At this point the only significant social change will probably occur when a sufficient amount of people,

through economic failure, lose confidence in their elected officials. The public will then demand other alternatives. While we would like to think that this could usher in a bright new chapter in the human drama, it is far more likely that the most probable course will be a form of dictatorship, perhaps even an American brand of fascism, ostensibly presented to the people as a way of protecting them from the products of their own inadequate culture.

However, it is not enough to point out the limiting factors that may threaten the survivability of all nations. The challenge all cultures now face in this technological age — some more than others — is to provide a smooth transition towards the introduction of a new way of thinking about ourselves, the environment and the management of human affairs.

It is now mandatory that all nations engage in a joint venture, planning on a global scale for new alternatives with a relevant orientation toward social arrangements. This is the only option if we are to avoid the unavoidable decline of the civilized world. If humankind is to come together toward a mutual prosperity, universal access to resources is essential.

Along with a new orientation toward human and environmental concerns, there must be the *methodology for making this a reality.* If these ends are to be achieved, the monetary system must eventually be surpassed by a world **resource-based economy.** To effectively and economically utilize resources, the necessary cybernated and computerized technology must be applied to ensure a higher standard of living for everyone. With the intelligent and humane application of science and technology, we will be able to guide and shape our future for the preservation of the environment, ourselves and for generations to come.

It is not enough to advocate the cooperation of all nations. We need a global society based upon a practical blueprint acceptable to the world's people. We also need an international planning council capable of translating the blueprint and the advantages that would be gained by world unification.

The design must be based upon the carrying capacity of our planet, its resources and the needs of its inhabitants. To sustain our civilization, we must coordinate advanced technology and available resources within a humane systems approach.

Many of the professions familiar to us today will eventually be phased out. With the rate of change now taking place, a vast array of occupations will become obsolete and disappear. In a society that applies a systems approach, these professions will be replaced by interdisciplinary teams — the systems analysts, computer programmers, operation researchers, and those who link the world together in vast communications networks.

We have the skills and the knowledge to apply interdisciplinary teams to problems. However, only in times of war or national emergencies do we call upon and assemble interdisciplinary teams to help find workable solutions to social problems. If we mobilize the same resources for our social problems as we do during a war, beneficial effects on a large scale can be achieved in a relatively short time. This could easily be accomplished by utilizing many of our universities' training facilities and staffs to best determine possible alternative methods to solve these problems.

This approach would be an important initial phase to define the possible parameters for the future of all civilization.

The process of social change must allow for changing conditions that continuously update the design parameters, and for the infusion of new technologies into emerging cultures. Design teams utilizing socially integrated computers can automatically be informed of any changes in conditions.

In this world of constant change it is no longer a question of whether we choose to make the necessary changes. Our very survival demands that we act on this challenge and adopt these new requirements.

CHAPTER 6

THE INHUMANITY OF A MONETARY-BASED SYSTEM

ALTHOUGH SKILLFUL ADVERTISERS LEAD US TO BELIEVE otherwise, in today's monetary-based economies, the human consequences of introducing new technologies seldom concern those introducing the technology. In a monetary-based system, the major aim of industry is profit: maintaining the competitive edge and the bottom line appears to be all that matters. The social and health problems that arise from mass unemployment of people, often of workers who are rendered obsolete by the infusion of automation, are considered irrelevant, if they are considered at all.

Any social need that may be met is secondary to acquiring a profit for the business. If the profit is insufficient, the service will be withdrawn. Everything is subordinate to increasing the profit margin for shareholders. It does not serve the interest of a monetary-based society to engage in the production of goods and services to enhance the lives of people as a goal — no more than manmade laws are enacted to protect the lives of its citizens.

All of the world's economic systems — socialism, communism, fascism, and even our free enterprise capitalist system — perpetuate social stratification, elitism, nationalism and racism, primarily based on economic disparity. As long as a social system uses money or barter, people and nations will seek differential advantage by maintaining the economic competitive edge or, if they cannot do so by means of commerce, by military intervention.

War represents the supreme failure of nations to resolve their differences. From a strictly pragmatic standpoint, it is the most inefficient waste of lives and resources ever conceived. This crude and violent attempt to resolve international differences takes on even more ominous overtones with the advent of elaborate computerized thermonuclear delivery systems, deadly diseases and chemicals, and the threat of sabotage to a nation's computer networks. Even when nations desire peace, they usually lack the knowledge to arrive at peaceful solutions.

War is not the only form of violence imposed on the populace by inadequate social arrangements. There is also hunger, poverty, and scarcity. The use of money and the creation of debt foster economic insecurity, a condition that perpetuates crime, lawlessness, and resentment. Paper proclamations and treaties do not alter the facts of scarcity and

insecurity, while nationalism tends only to help propagate the separation of nations and the world's people.

Even a peace treaty cannot prevent another war if the underlying causes are not addressed. The unworkable aspects of international law tend to freeze things as they are. Nations that have conquered land all over the world by force and violence would still retain their positions of territorial and resource advantage, regardless of treaties. Such agreements serve only as temporary suspensions to conflict.

But focusing our efforts on non-productive and non-creative endeavors wastes lives as surely as war does. Throughout history we have lived through ages characterized by wasted lives, in which the abilities of a great many people have not been fully realized of utilized. Countless time, efforts, and minds are wasted in the pursuit of money and on occupations that contribute nothing to the human intellect or condition.

From the earliest civilizations to the present day, most humans have had to work to earn a living. Our attitudes about work may be a carryover from earlier times.

During the thousands of years of the monetary system, most workers have been paid just enough to make it necessary that they return to work — even when higher wages have been possible. How else can the wage-payer keep the workers coming back? If the employee received wages that allowed them to work a few weeks and then take time off for a world cruise, an extended vacation, or other such luxuries, production schedules would suffer. Even the highly educated and affluent who live in more expensive homes and drive expensive cars have to appear at a place of work if they wish to maintain their standard of living. All of us — even the very top executives — are indentured slaves of the monetary system. Most of us suffer from the fact that we lack a meaningful existence. We stay at jobs we hate to buy more gadgets we don't need, or to build up earned time off so we can escape from the reason we need a vacation in the first place.

In the workaday world many of us are frantically busy trying to stay afloat, making payments on cars, homes, and material possessions that enslave the body and mind in an endless attempt to secure our future. Although many today take home much more money, the value of the dollar through inflation has decreased the purchasing power for most people. We are caught up in the game of getting ahead, without even thinking about what or whom we're trying to get ahead of. Most of us do not take the time to think about our own lives and how we relate to one another, or to what and who we really are.

Even those who achieve economic security have become addicted to the media's image of personal success. When we achieve our first economic goal, we want more— the cabin cruiser, the vacation house, and the trip abroad. In this monetary world even our dreams are rationed. We start out with "If only I can make a decent living". If we achieve that, we progress to "If only we had the little house in the country to get away to, then we'd be happy." At each successive gain in this endless chain of dissatisfaction we acquire more and more material wealth, but it's never enough to make us happy. We live in a world of unful-

filled dreams in which we never really come to know or understand what constitutes a meaningful life.

In the future people may view our present phase of civilization as an age of intellectual and economic insufficiency. They may find it difficult to understand how we could have accepted aggression and competition as being normal. Some mothers and fathers have even made attempts to secure the future for their daughters and sons by having them marry into a secure position of wealth. This is an accepted form of prostitution or selling to the highest bidder.

In a monetary system even democracy is an illusion perpetuated to give the populace a feeling of participation in a so-called democratic process. In most instances people nominated for public representation are pre-selected by the power elite and serve the interests of the highest bidder. Political parties are a perfect example: a single party representative runs against another party's single representative. The fantasy is that whichever one wins immediately represents everyone in the election area — regardless of political party or philosophy.

The country's actions and most of the decisions are made by and for the major corporations, financial interests, the most wealthy, and the military industrial complex. As long as money and a monetary system prevail, true democracy will continue to be nothing much more than an illusion.

We must stop constantly fighting to attain human rights and equal justice in an unjust system, and start building a society where equal rights are an *integral part of the design.*

As long as we remain in a monetary system, most people will never have the money to behave democratically. A person may desire a particular type of house and car, but might lack the means to purchase it. How has this person benefited from the democratic process or the freedom to choose? Yet we claim to have a democratic system that is the best form of government in the world. In actuality we are really only as free as our purchasing power enables us to be. And with money so concentrated in the hands of so few, even this freedom is illusory.

Despite its title and treatment by the media the Federal Reserve System that issues and controls our currency ***is not*** an agency of the federal government run for the benefit of people. Rather the "Fed" is a private institution run solely by and for private profit and even the amount in reserve is questionable.

The Federal Reserve, a private institution deceptively named, has enormous influence over our government, its leaders, our personal savings accounts, and to a large extent over how many of us will have jobs. The Federal Reserve, not the government, has complete control over the lending of money. This means it sets interest rates for our country, thereby maintaining tremendous political influence.

But the Federal Reserve System is not the only private institution that manipulates our economic system. Banks use a process called "fractional reserve banking" that enables them to loan much more money than they have assets on deposit, to cover the loans. They then charge interest on money they don't have. Through this process banks lend out at least

ten times more money than they have on reserve, which reduces the value of our money and leads to inflation. It is no wonder the newest and biggest buildings in all cities belong to banks. Of course, if we behave as the banks do we would be charged with fraud.

This is not a new practice. In 1881, James Garfield stated "Whoever controls the volume of money in any country is absolute master of all industry and commerce. And when you realize that the entire system is very easily controlled one way or another by a few powerful men at the top, you will not have to be told how periods of inflation and depression originate."

Private money lenders understood early on the overwhelming benefits of lending money to nations at war when the paybacks were assured by the taxation of their people. This was much more profitable than lending to individuals Financial interests and corporations instigate wars and disruptions to this very day.

Furthermore, the monetary system avoids the crises of the lack of purchasing power of individuals and small companies by propping up the economy through military expenditures, corporate welfare and providing funds for research to private industries. The government borrows money from private lending institutions to help support the economy in these areas. This increases the national debt while directing the public's attention away from national problems such as cutting expenditures from the veterans administration, aid to the poor, education, environmental concerns, etc. In many instances our government and corporations use our own military equipment and the use of force to put down revolutionary social changes elsewhere in the world, while generating an illusion of prosperity at home.

Amschel Rothschild, one of the early beneficiaries of the private banking system, stated, ***"Give me the power to issue and control a nation's money and I care not who makes its laws."*** *As it is applied today, financial power is truly amoral.*

A truly democratic system works only when all people have access to the same opportunities for individual development and economic growth. This requirement is not the goal of a monetary-based system.

In a free enterprise system, people who design and build a ski resort don't submit the design to a democratic vote in which all participate. Instead they submit it to the demands of the market, that is, those who can afford to ski. If they offer enough of the amenities that skiers want and can afford, their resort succeeds. A successful system should cater to the needs of all people. There are many people who would like to ski who can't afford to participate. Again, the choices are limited to what a certain group of people can afford. *This is elitism.*

Wherever money is involved, there is always a kind of elitism. Those who control the purchasing power always have a far greater influence.

Many years ago the American people were first taxed to build roads for automobiles. They did not vote for this development. The automobile and bus industries, real estate lobbies and military greatly influenced the development of freeways and roads because of the potential for automobiles and land sales represented by the expansion of the highway

system. Many cities offered transportation systems that were far cleaner, more efficient, and more economical than automobiles, but these were sold out and dismantled by the influence of the vested interests representing the automobile industry.

As a result, we have a transportation system that has resulted in urban sprawl, the loss of millions of acres of natural areas and croplands, air and water pollution, thousands of people killed or injured each year and transportation that is so expensive that a great number of Americans cannot afford to participate. For what and for whom has democracy worked in this case? Millions of Americans taxed for highways in which they do not share in the benefits and which have proved dangerous, inefficient and expensive as a means of transportation?

In our present monetary system private institutions hoard a great deal of useful knowledge, rather than making it readily available to the entire world's people. In an increasingly proprietary and commercial world, in which even college professors talk of copyrighting the notes from their lectures, there is a disturbing shift from the spirit of the pioneer to that of the salesman.

Several companies recently submitted and received patents on the genome of two men immune to AIDS. The companies neither created nor own the genetic material — nor did they discover them independent of the living bearers of those genes. But they acquired patents on genes in living humans. Is this democracy in action? At this writing, at least one of the men has threatened legal action. Rather than meeting the challenges of elevating the human condition, the scientist increasingly morphs into the businessman, auctioning the benefits of his or her efforts to the highest bidder. For this reason, a great deal of new technology still remains in the control of private institutions rather than the public domain.

Many of our heroes in the past have been honored for their self-sacrifice in attempting to make the world a better place. Thousands have sacrificed their lives; others have been tortured and imprisoned during their attempts to better the lives of others. Very often these people acted as they did without thought for monetary rewards.

The big lie that has been perpetuated by those in control of the money system is that only competition generates incentive. This system may be said to provide employment and incentive, but it also produces greed, corruption, crime, embezzlement etc. For centuries, all governments have directly and indirectly programmed their subjects with a value system that perpetuates the control group. They have successfully used the human mind as a dumping ground to store their own values and concepts along with behavioral patterns that generate feelings of guilt at any departure from the dominant values of the established system.

At the same time, these control groups have stifled the development of individuality by fostering compliant populations which lack both the information and the insight to pose the question, "Where, exactly, so my values come from?"

The monetary system places a tremendous, unnecessary strain on available resources, and denies access to the benefits of mass production to countless millions of people. In a

monetary-based society profit depends on maintaining an artificial scarcity of goods and services or on the conscious withdrawal of efficiency.

Rather than designing automotive styles to last for many years, manufacturers waste tremendous amounts of energy in retooling for yearly changes to compete for market share with others producing machines that serve exactly the same function. A recent military survey of commercial catalogs identified over 300 types of wrenches, differing so slightly that many were interchangeable. While a wrench is a useful tool, what purpose is served by over 300 minimal-varying models? Tremendous waste in materials and resources result from each company doing unnecessary and redundant paper work, advertising, manufacturing, etc.

A further example can be seen whenever a famous figure addresses the nation on television. The viewer will see dozens of different microphones, each representing a competing media-group. Actually only one or two are necessary to report the event throughout the world.

And consider the fashion industry, where clothing has to constantly change in order to induce people to purchase the newest, latest fads.

In the United States during periods of "price wars" milk and other agricultural products were destroyed to maintain higher price levels. Where is our outrage? We buy into the old "virtue" of work while we allow its essential products to be destroyed. Equally damaging, however, is planned obsolescence, where industries deliberately create products that break and require replacement or unnecessary repairs.

In the aircraft industry, sales of large transport aircraft are not the major source of profit. Large profits come from the maintenance and replacement of parts. This is particularly so within the military, with its dependence on the market. Vendor changes affect sustainment costs far more than changes in mission. Even during World War II, many anti-aircraft guns and armaments were manufactured from parts that were not interchangeable. The parts from one company would not fit the guns of another company.

Today Congress pushes the Defense Department to "save money" by purchasing commercial-off-the-shelf equipment. On the surface, buying already available equipment, rather than developing military-specific equipment, appears sensible. However, military equipment must be interoperable as well as supportable all over the world — and in environments that commercial equipment seldom encounters. Personal computers sent with troops to Desert storm, for example, failed by the hundreds in the excessive heat and sand encountered in Kuwait. And, instead of buying parts for a single model, the military now must accommodate multiple vendors and their equipment, parts and tools. Our tax money flows far outside government channels. This should surprise no one. Even a cursory review of the annual defense budget finds numerous examples of Congressionally-mandated purchases of equipment and services that are of no use to the military. So in addition to personally buying goods that keep companies in business, your taxes also go to them.

Over a half century ago the United States Electric Light Company gave its dynamic inventor Hiram Maxim a twenty thousand-dollar annual life pension and exiled him to

England. They needed to get rid of him because he kept inventing improvements. His creative ability made their equipment obsolete before they had the time to pay for it.

Unfortunately for the people in the U.S., Maxim produced some of his greatest inventions in England. At the same time he was being knighted for his outstanding accomplishments, the United States Electric Light Company was going out of business. In Japan today the shelf life of electronic equipment before obsolescence is approximately 3 months.

A money system has existed for centuries and, whether we realize it or not, has always been used to control the behavior of those with limited purchasing power. Money has no influence where resources and access are not limited. It is only when resources are scarce that a monetary or barter system can function.

In other words, if a person wants to obtain goods and services he or she is obligated to submit to the control of others. For example, when a person goes to work in any industry today, they enter a private dictatorship from the moment they punch the time clock to the time they leave the premises.

We are long past due for a serious examination and radical overhaul of our economic system and ideologies. Attempting to find solutions to the monumental problems within our present society will only serve as temporary patchwork, prolonging an obsolete system. Our competitive monetary system alone did not give us a high standard of living in the United States. Our advantages also come from being isolated from hostile, neighboring countries, from our wide range of natural resources, our fertile land and the many contributions of inventors and engineers, and from our production technology.

CHAPTER 7

WHEN MONEY BECOMES IRRELEVANT

IN THIS CHAPTER WE WILL DISCUSS a straightforward approach to the redesign of a culture, in which the age-old inadequacies of war, poverty, hunger, debt, and unnecessary human suffering are viewed not only as avoidable, but totally unacceptable. This new social design works towards eliminating the underlying causes that are responsible for many of our problems, but they cannot be eliminated within the framework of the present monetary and political establishment.

The major corporations' first concern is profit. This narrow vision will ultimately result in the demise of our economic system. If the monetary system continues to operate, we face ever-increasing technological unemployment, today referred to as downsizing. We need fewer people with greater skills to support our production. Everyone else becomes irrelevant — except as consumers of that production. It is only a matter of time until automation replaces people in almost all areas, resulting in the lack of purchasing power for people to buy the goods turned out. Even with an expanding market, such a situation will inevitably bring about massive and unmanageable problems.

During the 1930's, at the height of the Great Depression, the Roosevelt administration enacted new social legislation designed to minimize revolutionary tendencies and to address the problems of unemployment by providing jobs through the WPA, CC Camps, NRA, transient camps, and the federal arts projects. Ultimately, however, it was World War II that pulled the U.S. out of that worldwide recession. If we permit current conditions to take their natural course, we will soon be faced with another international recession of even greater magnitude.

Although at the beginning of World War II, the US had only 600 first class fighting aircraft, we rapidly increased production to 90,000 planes per year. Did we have enough money to pay for the required implements of war? No, there was not enough money or gold, but we did have more than enough resources. Available resources and personnel — not funding — provided production and efficiency required to win the war.

We live in a culture that seems to work collectively only in response to a crisis. Only in times of war or national disaster do we move beyond money and apply the necessary

resources and assemble interdisciplinary teams to meet a threat. Rarely, if ever, do we employ such concerted efforts to find workable solutions to social problems.

If we applied the same efforts of scientific mobilization toward social betterment as we do during a war or disaster, large-scale results could be achieved in a relatively short time. Much more time and effort must be directed toward the collection of experimental evidence to support innovative social arrangements.

The Earth is abundant with resources; today our practice of rationing resources through monetary methods is irrelevant and counterproductive to the well-being of people. Modern society has access to highly advanced technologies and can easily provide more than enough for a very high standard of living for all the earth's people by implementing a **resource-based economy.**

Simply stated, a resource-based economy utilizes existing resources rather than money, and provides an equitable method of distribution in the most humane and efficient manner for the entire population. It is a system in which all natural, man-made, machine-made, and synthetic resources are available without the use of money, credits, barter, or any other form of debt. A resource-based economy utilizes existing resources from the land and sea, physical equipment, industrial plants, etc. to enhance the lives of the total population. In an economy based on resources rather than money, we can easily produce all of the necessities of life and provide a high standard of living for all.

In a resource-based economy all of the world's resources are held as the common heritage of all of the earth's people, thus eventually outgrowing the need for the artificial boundaries that separate people. ***This is the unifying imperative.***

We must emphasize here that this approach to global governance has nothing whatever in common with the present aims of an elite to form a world government with themselves and large corporations at the helm, and the vast majority of the world's population subservient to them. Our vision of globalization empowers each and every person on the planet to be the best they can be, not to live in abject subjugation to a corporate governing body.

All social systems, regardless of political philosophy, religious beliefs, or social customs, ultimately depend upon natural resources, i.e. clean air and water, arable land and the necessary technology and personnel to maintain a high standard of living. This can be accomplished through the intelligent and humane application of science and technology. The real wealth of any nation lies in its developed and potential resources and the people who work toward the elimination of scarcity and the creation of a more humane society.

The utilization of large-scale computer based systems will assist us in defining the possible parameters required for the operation of a resource-based economy, and all construction projects would conform to environmental requirements. Over-exploitation would be unnecessary and surpassed. As of this writing NASA has announced the use of a powerful, parallel supercomputer to evaluate the global impact of national and human-induced activities on our climate.

A resource-based economy would use technology to overcome scarce resources and utilize renewable sources of energy, to computerize and automate manufacturing and inventory, and to design safe, energy-efficient cities while providing universal health care and better education. Equally importantly, it would generate a new incentive system based on human and environmental concerns.

Unfortunately science and technology have been diverted from these ends for reasons of self-interest and monetary gain through planned obsolescence, sometimes referred to as the conscious withdrawal of efficiency. For example, it is an ironic state of affairs when the U. S. Department of Agriculture, whose function is to conduct research into ways of achieving higher crop yields per acre, actually pays farmers not to produce at full capacity while many people go hungry.

In addition, we place signs next to a highway that read "Caution: Slippery When Wet," when a more effective approach would be to design a road with abrasive strips that would not be slippery. We continue to "purify" our water systems by dumping chemicals into them, despite their continuing build-up. Then again, there is the problem of the dumping of waste into rivers and waterways because it is cheaper than other, more responsible disposal methods. Yet another example is the failure of most industry to install electrostatic precipitators at their plants to prevent particulate matter from being released into the atmosphere from industrial smokestacks, a technology that has been available for over 75 years. The monetary system does not always apply known methods that would best serve people and the environment.

In our search for more, we have blinded ourselves to our personal responsibility for challenging these absurdities. A resource-based society considers us all equal shareholders of the earth who are responsible both for it and for our relationship with each other.

In a resource-based economy, the human condition is of prime concern with technology subordinate. In such an economy, production is accomplished totally by machines and the products are available to all. The concepts of "work" and "earning a living" become irrelevant. The focus is on making a life. In a resource-based economy, if the human consequences of automation are neglected, it renders all the advances of science and technology meaningless except to a selected few.

To better understand the meaning of a resource-based economy consider this: if all the money in the world suddenly disappeared, as long as topsoil, factories, and other resources were left intact, we could build anything we chose to build and fulfill any human need. *It is not money that people need, but rather it is freedom of access to the necessities of life without having to appeal to a government bureaucracy or any other agency.* In a resource-based economy money will be irrelevant. All that would be required are the resources, and the manufacturing and distribution of the products.

In a monetary system, purchasing power is not related to our capacity to produce goods and services. For example, in a recession there are computers in store windows and automobiles in car lots, but most people do not have the purchasing power to buy them.

The rules of the monetary system are obsolete and create needless strife, deprivation, and human suffering.

In today's culture of profit, we do not produce goods based on human need. We do not build houses based on population needs. We do not grow food based solely on demand, nor do we practice medicine solely to cure diseases. Industry's major motivation is profit.

Take the automobile. To service conventional automobiles today we have to remove a great deal of hardware before we can get to the engine. Why are they so complicated? Simply because ease of repair is not the concern of the manufacturers. They do not have to pay to service the car. An entire subset of the automotive industry is devoted to making a profit from the repair of cars and trucks. If manufacturers were responsible for the cost of repairs, cars and trucks would be built very differently — and with different materials and performance — with modular components easily disengaged to get at the engine.

Such construction would be typical in a resource-based economy. Many of the components in the automobile would be easily detachable to save time and energy in the rare case of repair, because no one profits by servicing automobiles or any other products. Consequently quality and simplicity of servicing and technological upgrade would be primary design drivers. Eventually, with the development of magnetically suspended bearings, lubrication and wear could be relegated to the past. There will also be proximity devices on all vehicles to prevent collisions. Automotive transport units engineered in this way would easily be service-free for many years.

This same process would be applied to all products. All industrial devices would be designed for recycling; but there would be much less recycling when we build household materials and products of superior quality that are designed not to wear out or break down.

A resource-based world economy would also involve all-out efforts to develop new, clean, and renewable sources of energy: geothermal; controlled fusion; solar; photovoltaic; wind, wave, and tidal power; and even fuel from the oceans. We would eventually be able to have energy in unlimited quantity that could propel civilization for thousands of years. A resource-based economy must also be committed to the redesign of our cities, transportation systems, and industrial plants, allowing them to be energy efficient, clean, and conveniently serve the needs of all people.

All shipping and transportation systems would maintain a balanced load economy, fully utilized in both directions. There would be no empty trucks, trains, or transport units on return trips. There would be no freight trains stored in yards dependent on the business cycle for their use.

What else would a resource-based economy mean? Technology intelligently and efficiently applied, conserves energy, reduces waste, and provides more leisure time. With automated inventory on a global scale we can maintain a balance between production and distribution. Planned obsolescence would be unnecessary and non-existent in a resource-based economy. All packaging systems would be standardized, requiring less storage space and easy handling. Only nutritious and healthy food would be available. To eliminate waste from

impermanent products, such as newsprint, books and magazines, a light-sensitive film can be placed over a monitor or TV to produce a temporary printout of the news of the day or any other relevant information. This material will be capable of holding the information until it is deleted. This would conserve millions of pounds of paper, a major part of the recycling process. Eventually most of the paperwork, including the transfer of money, would be eliminated.

As we outgrow the need for professions based on the monetary system, for instance those of lawyer, banker, insurance agent, advertiser, salesperson, and stockbroker, a considerable amount of waste will be eliminated. Considerable amount of energy would also be saved by eliminating the duplication of competitive products such as tools, eating utensils, pots and pans, vacuum cleaners and those 300 wrenches mentioned earlier. Choice is good. But instead of hundreds of different manufacturing plants and all the paperwork and personnel required to turn out similar products, only a few of the highest quality would be needed to serve the entire population.

Our only shortage is the lack of creative thought and intelligence in ourselves and our elected leaders to solve these problems. The most valuable, untapped resource today is human ingenuity. With the elimination of debt, the fear of losing one's job will no longer be a threat This assurance, combined with education on how to relate to people in a much more meaningful way, could considerably reduce both mental and physical stress and leave us free to explore and develop our abilities.

If the thought of eliminating money still troubles you, consider this: If a group of people with gold, demons and money were stranded on an island that had no resources, their wealth would be irrelevant to their survival. It is only when resources are scarce that money can be used to control their distribution. One could not, for example, sell the air we breathe or water abundantly flowing down from a mountain stream. Although air and water are valuable, in abundance they cannot be sold.

Money is only important in a society when certain resources for survival must be rationed and the people accept money as an exchange medium for the scarce resources. Money is a social convention, an agreement if you will. It is neither a natural resource nor does it represent one. It is not necessary for survival unless we have been conditioned to accept it as such.

WHAT WILL MOTIVATE PEOPLE?

Some people claim the free-enterprise system and its competition create incentive. This is only partially true. It also perpetuates greed, embezzlement, corruption, crime, stress, and economic hardship and insecurity. Most of our major developments in science and technology resulted from the efforts of very few individuals, working independently and often against great opposition: Goddard, Galileo, Darwin, Tesla, Edison, Einstein, etc. These individuals were also genuinely concerned with solving problems and improving processes,

rather than with mere financial gain. Despite our slavish belief that money is incentive, we often mistrust those whose sole motivation is monetary gain. This can be said for doctors, lawyers, and those in just about any field.

If the basic necessities are accessible, what will motivate us? Quite simply, the things we care about. Children reared in affluent environments, in which food, clothing, shelter, nutrition, extensive education and much more are provided, still demonstrate incentive and initiative. On the other hand, there is overwhelming evidence to support the fact that malnutrition, lack of employment, minimum wage, poor health, a lack of direction, lack of education, homelessness, little or no reinforcement for one's efforts, poor role models, poverty, and a bleak prospect for the future can and do destroy incentive and the drive to achieve.

The aim of our new social design is to encourage a new incentive system, one that is no longer directed toward the shallow and self-centered goals of wealth, property, and power. These new incentives would encourage people toward self-fulfillment and creativity, the elimination of scarcity, the protection of the environment, and most of all concern for their fellow human beings.

The air we breathe, clean water, sunshine, forests, and the totality of nature for the most part support life without charge. Provided with good nutrition in a highly productive humane society, people will evolve a new incentive system. Without the need to work to survive, there would be so many new things to explore and invent that the notion of people sitting around and doing nothing is just absurd. The lack of incentive in our present culture occurs when people dare not dream of a future that seems unattainable to them.

Each phase of social evolution creates its own incentive system. In primitive times the incentive to hunt for food was generated by hunger; the incentive to create a javelin or a bow and arrow evolved as a process supportive to the hunt. With the advent of the agrarian revolution the motivation for hunting was no longer relevant, and incentives shifted to the cultivation of crops and supporting implements, the domestication of animals, and protection of personal property. In a civilization where people receive food, medical care, education, and housing, incentives would again change appropriately. People would be free to explore other possibilities and lifestyles that could not be anticipated in earlier times.

The nature of incentive and motivation depends on many factors. We know, for example, that an individual's physical and mental health directly relates to that person's level of motivation and productivity, as we are also aware of the fact that all healthy babies are naturally inquisitive. In India and other areas of great and continuing scarcity many people are motivated against the accumulation of wealth and material property. So, they renounce all worldly goods. Under those conditions this is not difficult. This would seem to be in direct conflict with western culture's emphasis on the accumulation of material wealth. Yet, which is more valid? Your answer to this question would depend upon your culturally influenced value-system.

Some people overcome the shortcomings of their environment in spite of an apparent lack of positive reinforcements. They provide their own "self-reinforcement", are able to see an improvement in whatever it is they are engaged in, and achieve an intrinsic sense of accomplishment. Their reinforcement does not depend on the approval of others. Those children who do depend on the approval of a group tend to have low self-esteem. Children who do not depend on group approval acquire a sense of self-esteem by improving upon their performance.

Throughout history, many innovators, artists, and inventors have been ruthlessly exploited, ridiculed, and abused while receiving very little financial incentive. Yet, they endured such hardship because they were motivated to learn and to discover new ways of doing things. On the other hand, Leonardo da Vinci, Michelangelo, and Beethoven, to name a few of history's most creative minds, received the generous sponsorship of wealthy patrons. Yet this did not kill their incentive in the least; on the contrary, these individuals strove to reach new heights of creativity, perseverance and individual accomplishment. Creativity is often its own motivation

This is a difficult concept to grasp because most of us have been brought up with a set of notions about the way that we ought to think and behave. These are based upon ancient ideas that are actually irrelevant today. To a limited extent, some primitive cultures in remote islands in the South Pacific have full access to all the food they require and to clean water and air. There is probably no question that most of them are far better adjusted than many people in the so-called civilized world. There is no evidence that demonstrates that unlimited access to the necessities of life diminishes incentive.

It has often been stated that war spurs creativity. This deliberately falsified concept has no basis in fact. It is the increased government financing of the war industries that helps develop so many new materials and inventions. There is no question a saner society would be able to create a more constructive incentive system if our knowledge of the factors that shape human motivation were applied.

Experimental psychologists have shown that the environment plays a major role in shaping our behavior and values. If constructive behavior is appropriately rewarded during early childhood, the child becomes motivated to repeat the rewarded behavior, provided the reinforcement meets the individual needs of the child. For example, a football given to a child who is interested in botany would not be a reward from the child's point of view. It is very unfortunate that so many individuals in our society today are not appropriately rewarded for their creative behavior.

In a resource-based economy, motivation and incentive will be encouraged through recognition of, and concern for, the needs of the individual. This means providing the necessary environment, educational facilities, good nutrition, health care, warmth, love, and security that people require.

LOVE AND EXTENSIONALITY

For centuries love has been a dominant word in our vocabulary. The definition varies so much today that the word has become almost meaningless. Love is subject to many different interpretations, most of them irrelevant to the behavior associated with it. Perhaps the word love may one day be redefined in much more relevant terms, such as our being *extensional* to one another. What is being extensional? Our arms and hands enable us to pick up and rotate objects, to view them from many different positions. Our arms and hands are extensional devices along with our eyes, ears, nose and our entire physical body.

When a single individual tries to build a log cabin, it may take considerable time to complete. With the help of several neighbors this job can be completed in a relatively short time. The neighbors become very extensional to this individual. The same is true with an entire community that acts in a supportive manner towards one another.

In the physical sciences a structural engineer has to work with the metallurgist to help improve the strength and quality of structural materials. The two and their skills, their teamwork, are genuinely extensional. The physical sciences are the closet approach to genuine extensionality. Rather than being directed at a single individual, genuine extensionality serves all people equally. For instance, when contaminated water is purified, it benefits any person that utilizes the process. Vaccinating children to prevent disease is not only extensional to the individual but to almost everyone that they come in contact with. Identifying conditions responsible for a disease is extensional and beneficial to all people regardless of their personal values and philosophy.

When different nations share their technology intelligently, it is extensional to all people regardless of their independent beliefs and national interests. Corporate systems, however, chiefly benefit the owners and the shareholders. When inventions serve the needs of all people they will truly be extensional.

To know the difference between governments and people that merely verbalize good intentions as opposed to displaying actual extensionality is essential to advancing civilization both physically and intellectually.

When a bank lends money to an individual there is a benefit from this but this comes at a cost called debt and obligation. Genuine extensionality does not extract a toll. Extensionality at its most basic is a simple act of kindness that one does without any debt being incurred by the other person.

The more people become extensional to one another, the richer the civilization and the interaction between individuals becomes. In the future, instead of asking "Do I love this person?" one might instead identify the specific areas of extensionality that one shares with them.

WOULD ALL THE PEOPLE IN SUCH A CYBERNATED SOCIETY BE UNIFORM?

Yes, in many ways, they would. For example, everyone would possess an understanding of the importance of extending maximum courtesy to all nations and to one another. They would share an intense curiosity for all that is new and challenging. With a better understanding people could possess a flexibility of outlook unknown in previous times, free of bigotry and prejudice. In addition, the people of this new society would care for their fellow human beings and for the protection, maintenance, and stewardship of the Earth's natural environment. Additionally, everyone — regardless of race, color, or creed — would have equal access to all of the amenities that this innovative culture could supply.

SOMETHING FOR NOTHING

Some people question the morality of receiving something for nothing. Once when I was speaking at a college a student expressed opposition to the idea of getting something for nothing. I asked if he would answer a personal question and when he replied, yes, I asked him "Are you paying your way through school or are you parents?" He admitted his parents were. I pointed out that if he really did believe people should not receive something for nothing then in the event of the death of a family member he would prefer the inheritance be left to the heart or cancer fund, rather than passing to him. Needless to say, the students were opposed to this idea.

Merely being born in a developed country, we have access to many things we put no effort toward, such as the telephone, the automobile, electricity, and running water. These gifts of human ingenuity and invention do not degrade our lives, but rather enrich us. What degrades us is our lack of concern for those unfortunate enough to experience poverty, hunger, lack of medical care, and war. The social designs that are proposed in this writing merely provide the opportunity for individuals to develop their fullest potential in whatever endeavor they choose without the fear of any loss of individuality.

WHAT GUARANTIES PEOPLE THE RIGHT OF PARTICIPATION?

There is always a reason for corruption and that is that someone gets something they consider valuable out of it. Without vested interests or the use of money, there is little to be gained by squelching opinion, falsifying information or taking advantage of anyone. There are no underlying, rigid social barriers to limit the participation of anyone or restrain the introduction of new ideas. The main objective is full access to information and the delivering of goods and services to all people, a state of affairs that will enable people to be prepared to participate in the exciting challenges of this new society.

HOW WOULD RESOURCES BE DISTRIBUTED EQUITABLY IN A RESOURCE-BASED SOCIETY?

Distribution of goods and services without the use of money or tokens would be accomplished through the establishment of distribution centers. These distribution centers would be similar to expositions, where the advantages of new products could be explained and demonstrated. For example, if you visited Yellowstone National Park you could check out a camera or camcorder, use it and then return it to another distribution center or drop-off, eliminating storing and maintenance.

Besides the computerized centers throughout the various communities where products would eventually be displayed, there will be 3-D, flat-screen imaging capabilities in each home. If you desire an item, an order can be placed and the item automatically delivered directly to your place of residence. All raw materials for the manufacture of these products can be transported directly to the manufacturing facilities by automated transportation "sequences" such as boats, monorails, mag-lev trains, pipelines, and pneumatic tubes. An automated inventory system would be connected to both the distribution centers and the manufacturing facilities, thus coordinating production to meet demand and providing a constant evaluation of preferences and consumption. In this way a balanced-load economy can be maintained. Shortages, over-runs, and waste could be eliminated.

IN CONCLUSION

Despite today's National Security mania for intruding on everyone's privacy, in a resource-based economy no one need take from another. It will be considered socially offensive and counterproductive for machines to monitor the activities of human beings, but more to the point, there will be no reason for it. The main concern of this new social arrangement is to create an environment that will encourage the widest range of individuality, creativity, constructive endeavor and cooperation without any kind of elitism, technical or otherwise. Most significantly, a resource-based economy would generate a far different incentive system, one based on human and environmental concern. This would not be a uniform culture but one that is designed to be in a constant process of growth and improvement.

It also advocates the stabilizing of the world's population through education, until such a point as the human population coincides with the earth's carrying capacity. Only when population exceeds the productive capacity of the land do many problems such as greed, crime, and violence emerge.

As we enhance the lives of others, protect our environment, and work toward abundance, all our lives can become richer and more secure. If these values were put into practice, it would enable all of us to achieve a much higher standard of living within a relatively short period of time — a standard of living that would be continuously improved.

In the society of the future, when the monetary system and scarcity are replaced by a resource-based economy and most of our needs are met, private ownership as we know it

would cease. The concept of ownership would be of no advantage whatsoever in a high-energy society.

Although this is difficult for many to imagine, even the wealthiest person today would be immensely better off in a high-energy, resource-based society. Today the middle classes live better than kings of times past. In a resource-based economy everyone would live far better than the powerful and wealthy of today.

People would be free to pursue whatever constructive field of endeavor they chose, without any of the economic pressures, restraints, and taxation that are inherent in the monetary system. By constructive endeavor, we mean anything that enhances the lives of the individual and others. When education and resources are available to all without a price tag, there will be no limit to the human potential. With these major alterations people would eventually be able to live longer, more meaningful and healthier lives. The measure of success would be based on the fulfillment of one's individual pursuits rather than on the acquisition of wealth, property, and power.

This proposal is neither Utopian nor Orwellian, nor does it reflect the dreams of impractical idealists. Instead, it offers attainable goals requiring only the intelligent application of what we already know. The only limitations are those which we impose upon ourselves.